

James, 40 · two focal seizures in one week (right hand twitching, lip-smacking) · subtle right hand weakness · MRI: **2-cm enhancing mass in the left frontal lobe** · biopsy pending · **likely glioma**

BIO 004 · WEEK 7 · CASE DEEP DIVE

The Brain

A 40-year-old who hadn't had a seizure until last week. A mass in his left frontal lobe. Every symptom localizes.

PATIENT CASE

James is 40. A week ago his wife noticed his **right hand twitching** while he was watching TV; he was awake but couldn't stop it for about 90 seconds. Yesterday he had a second episode and his speech became briefly garbled. No prior seizure history.

Neurologic exam reveals **subtle right hand weakness** and slight word-finding difficulty. MRI brain with contrast shows a **2-cm ring-enhancing mass in the left frontal lobe**, near the precentral gyrus. He is started on an antiepileptic and steroids; neurosurgery is consulted for biopsy.

Every one of James's symptoms localizes to a specific region of his brain. The brain is the most regionally specialized organ in the body — and tumor symptoms read its anatomy.

GUIDING QUESTION

How are the cerebral hemispheres organized, what does each lobe do, and how does a focal lesion produce a focal deficit that lets you map back to the tumor?

PREWORK

Companion to your Week 7 Brain prework page.

Two halves connected by the corpus callosum

The cerebrum has two **cerebral hemispheres** connected by the **corpus callosum** (a massive white-matter bundle, ~200 million axons). Each hemisphere has four major lobes plus the insula (buried in the lateral fissure).

The hemispheres are largely mirror images, but with important asymmetries: language areas (Broca's, Wernicke's) are typically left-hemisphere in 95%+ of right-handed people; spatial/musical processing is more right-hemisphere.

Four major regions with distinct jobs

Frontal lobe

Front of brain. Primary motor cortex (precentral gyrus), premotor cortex, Broca's area (dominant hemisphere), prefrontal cortex (executive function, judgment, personality). *James's tumor location*.

Parietal lobe

Top/back. Primary somatosensory cortex (postcentral gyrus), spatial awareness, body schema. Damage causes neglect, apraxia.

Temporal lobe

Sides. Primary auditory cortex, Wernicke's area (language comprehension), hippocampus (memory), amygdala (emotion).

Occipital lobe

Back. Primary visual cortex (calcarine fissure area). Damage produces visual field deficits.

Insula

Hidden in the lateral fissure. Visceral sensation, taste, interoception.

The homunculus

The **primary motor cortex** (precentral gyrus) and the **primary somatosensory cortex** (postcentral gyrus) are organized as *topographic maps* of the body. Areas with fine control (hands, face, lips, tongue) take up a disproportionately large area; areas with cruder control (trunk, leg) take a smaller share. The famous "homunculus" diagram exaggerates body parts according to cortical real estate.

The leg representation is on the medial surface (interhemispheric fissure). The arm and face are on the lateral surface. James's tumor near the left precentral gyrus would predict right hand and face symptoms — which is exactly what he has.

Basal ganglia and limbic system

● Basal ganglia

Caudate, putamen, globus pallidus + substantia nigra, subthalamic nucleus. Modulate movement.
Damaged in Parkinson's (substantia nigra),
Huntington's (caudate), hemiballismus (subthalamic).

● Limbic system

Hippocampus, amygdala, cingulate gyrus, fornix, mammillary bodies. Emotion, memory, motivation.
Hippocampal damage = anterograde amnesia.

Thalamus, hypothalamus, epithalamus

Thalamus: nearly all sensory information (except smell) relays through here on its way to cortex. Multiple specialized nuclei. Often called the "switchboard of the brain."

Hypothalamus: floor of the diencephalon. Coordinates autonomic, endocrine, and behavioral homeostasis (temperature, thirst, hunger, circadian rhythm, sex drive). Controls pituitary.

Epithalamus: includes pineal gland.

Two pairs of arteries, one circle

Brain blood supply comes from two pairs of arteries: **internal carotid arteries** (anteriorly) and **vertebral arteries** (posteriorly). They join at the base of the brain to form the **circle of Willis**, which gives off the major cerebral arteries: anterior cerebral (medial frontal/parietal), middle cerebral (lateral surfaces), posterior cerebral (occipital + inferior temporal).

The circle provides collateral flow: if one inflow is occluded, the others can sometimes compensate. Common stroke patterns map directly to which artery is occluded.



DRAW ZONE 1 -- Sketch the lateral surface of the left cerebral hemisphere. Label the four lobes, central sulcus, lateral fissure, primary motor cortex, primary sensory cortex, Broca's area. Mark James's tumor.

Tumor is in the left frontal lobe just anterior to the precentral gyrus. Show how it could affect hand area and Broca's area.



DRAW ZONE 2 -- Draw the motor and sensory homunculi on the precentral and postcentral gyri respectively. Exaggerate hands, face, lips.

Leg representation is on the medial surface (draw a sagittal view alongside).

RETURNING TO JAMES

A tumor that announces itself in body parts

James's symptoms localize because the brain is regionally specialized. Every finding maps to the location of the tumor.

"Focal motor seizures (right hand twitching)"

Tumor irritates the left motor cortex hand area. Seizure begins focally as the hand representation depolarizes.

"Word-finding difficulty during seizure"

Seizure activity spreads from motor cortex into nearby Broca's area (also left frontal in dominant hemisphere). Expressive aphasia.

"Subtle right hand weakness between seizures"

Tumor compresses or invades left motor cortex hand area. Persistent (interictal) weakness layered onto episodic (ictal) seizure activity.

"Why focal seizures vs generalized"

Tumor is a focal lesion. Seizure activity arises from the irritated cortex and (in James's case) hasn't generalized to involve the whole brain (which would cause loss of consciousness).

"Ring enhancement on MRI"

Enhancing rim indicates a region with disrupted blood-brain barrier (contrast leaks in). Central non-enhancement often = necrosis. Typical of high-grade glioma, also seen in abscess or metastasis.

"Why steroids"

Dexamethasone reduces vasogenic edema around the tumor, decreasing mass effect and symptoms. Buys time before definitive treatment.

Lobe, gyrus, homunculus, artery. James's tumor is teaching brain localization in real time.

